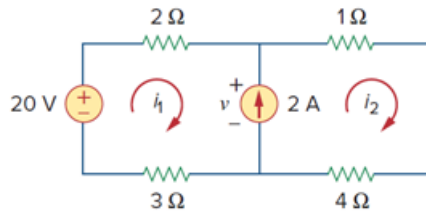


Problem

The voltage v across the current source in the circuit of Fig. 3.49 is:

- (a) 20 V
- (b) 15 V
- (c) 10 V
- (d) 5 V

Figure 3.49:



Step-by-step solution

Step 1 of 3

Refer to Figure 3.49 in the textbook.

From the current source, the relation between the currents is,

$$i_2 - i_1 = 2$$

$$i_2 = 2 + i_1$$

Step 2 of 3

Apply Kirchhoff's voltage law to the overall loop.

$$-20 + 2i_1 + 1i_2 + 4i_2 + 3i_1 = 0$$

$$5i_1 + 5i_2 = 20$$

Substitute $2 + i_1$ for i_2 .

$$5i_1 + 5(2 + i_1) = 20$$

$$5i_1 + 5i_1 + 10 = 20$$

$$10i_1 = 10$$

$$i_1 = 1 \text{ A}$$

Step 3 of 3

Apply Kirchhoff's voltage law to loop 1.

$$-20 + 2i_1 + v + 3i_1 = 0$$

$$v = 20 - 5i_1$$

Substitute 1 A for i_1 .

$$v = 20 - 5(1)$$

$$= 20 - 5$$

$$= 15 \text{ V}$$

Thus, the correct option is **(b)**.